

Exploring How, for Whom and in Which Contexts Extended Reality Training ‘Works’ in Upskilling Healthcare Workers: A Realist Review

Norina Gasteiger*, Sabine N van der Veer, Paul Wilson and Dawn Dowding

School of Health Sciences, University of Manchester

ABSTRACT

Extended reality (XR), including virtual reality (VR) and augmented reality (AR) may overcome barriers to training healthcare workers, such as resource constraints. However, the effectiveness of XR training is disputed and not well understood. Our realist review explores how, for whom and in what contexts AR and VR training ‘works’ in upskilling healthcare workers. Eighty papers informed our program theory, while 46 empirical studies tested/refined it. We conclude that XR triggers perceptions of realism and deep immersion, and enables visualization, interactive learning, enhancement of skills and repeated practice within a safe learning environment, consequently bettering skills, learning/knowledge and learner satisfaction.

Keywords: Healthcare, virtual reality, training, augmented reality

Index Terms: J.3 [Computing Applications]: Life and Medical sciences; J.4 [Computing Applications]: Social and Behavioral Sciences

1 INTRODUCTION

Within health and care settings, upskilling is required for promoting workforce flexibility and implementing best practice. This requires learning new skills or refining existing skillsets. Traditionally, this may include shadowing, the ‘see one, do one’ method, textbooks, scientific literature, e-learning, workshops and simulation-based methods (e.g., use of cadavers, live animals, patient actors, bench-top simulators or manikins). However, reduced training time and resources, costs, commitment to patient safety and ethical concerns have created barriers to learning via simulation [1].

Extended reality (XR) tools such as virtual reality (VR) and augmented reality (AR) might overcome these challenges. VR refers to a computer-generated simulated environment that immerses users. This pertains to non-immersive content delivered on computer screens to fully immersive VR accessed through headsets and/or haptics for force feedback and tactile sensation. In AR, digital imagery complements the real-world environment.

However, the efficacy of XR training is complex as it relies on perceptions of fidelity (realism). This depends on open-mindedness, psychological factors (e.g., stress or emotions) and realistic physical environments, functionality, psychological responses and social interactions (e.g., in multi-user games) [2, 3]. XR programs do also not always ‘work’ equally for everyone [4, 5], while implementation considerations are scarcely reported.

In this realist review we have developed, tested and refined a program theory that explores how, for whom and to what extent AR or VR training ‘works’ for upskilling healthcare workers. We also aimed to understand what facilitates/constrains implementation. The full-text paper is available elsewhere [6].

2 METHODOLOGY AND METHODS

We conducted a theory-driven realist review to help explain why there is variation in the success of XR training for health workers. Realist reviews consider causal explanations of how mechanisms might be triggered, that then lead to intended or unintended outcomes [7]. There is, therefore, an expectation that XR training does not lead to outcomes alone, but instead depends on mechanisms such as changes in reasoning or reaction (e.g., behaviors, perceptions) [7]. These are triggered in specific circumstances, contexts or conditions. By considering contexts, mechanisms and outcomes, a program theory consisting of CMO configurations (statements) is developed. The configurations are stated as: **Context + Mechanism/s = Outcome/s**. Our review was completed in 3 steps: theory elicitation, testing and refinement [8].

2.1 Theory Elicitation

A total of 7 databases and 11 journals were searched for literature on XR training for registered health and care workers. Literature also reported detail on contexts, mechanisms and outcomes. We screened 948 abstracts and 186 full-texts for eligibility, of which 80 papers were included. A narrative synthesis was conducted on the extracted information relating to contexts, mechanisms, and outcomes, to elicit an initial theory.

2.2 Theory Testing

Empirical studies were identified through another search of the databases. We reviewed 44 abstracts and 7 full-text papers for eligibility. 44 empirical studies from Step 1 were also included. A total of 46 empirical studies were used to test the theory, by providing evidence for the mechanisms and outcomes. The established Mixed Methods Appraisal Tool (MMAT) [9] was used to assess the quality of the research.

2.3 Theory Refinement

To refine the theory, evidence from Step 2 was compared to the initial program theory (Step 1) and revisions were made from any differences. The final theory was presented as a narrative and diagrammatic summary. Confidence in the CMO configurations (very low, low, moderate or high) was assessed using MMAT scores, contesting evidence and the number of supporting studies.

3 RESULTS

Of the 80 papers used to form the initial theory, most ($n=66$) used VR, with only 9 reporting on AR and 5 on both. The 46 empirical studies were conducted globally (e.g., America, Europe, China, Australia, India and Taiwan) and were published between 1999-2021. The participants had between 2 months and 30 years of experience and were from various health and care roles, including nurses, dentists, social workers, doctors, surgeons, pharmacists, technicians and respiratory therapists. The research was mostly moderate ($n=30$) or high quality ($n=13$). Only 3 were low quality.

* norina.gasteiger@postgrad.manchester.ac.uk

The programs attempted to improve healthcare workers' technical behavioral skills (e.g., dental drilling and emergency evacuation) or non-technical skills (e.g., decision-making or team-work). Non-immersive VR was used in 22 studies (e.g., LapSim) while 12 used fully immersive VR (e.g., Oculus Rift). Two used the CrystalEyes shutter glasses for partially immersive VR and six used AR (e.g. Microsoft HoloLens glasses). Haptics were used in 24 studies and other tools included manikins, actors and the da Vinci robot. For example, a simulator (Fig. 1) prompted actors to give patient feedback (groaning noises) in endoscopy training [10].



Fig 1: Actors and non-immersive VR for endoscopy training [10]; HoloLens AR headset and manikin for CPR training [11].

3.1 Context-Mechanism-Outcome Configurations

We had moderate or high confidence in nine of the 41 configurations (diagrammatic summary is shown in Fig 2).

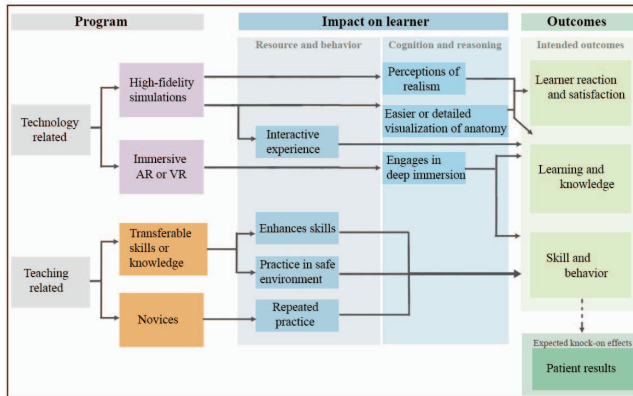


Fig 2: Diagram showing CMO configurations with most confidence.

We found that when AR or VR portray high fidelity simulations then perceptions of realism, easier visualization (i.e., of patient anatomy) and an interactive learning experience are triggered. This may result in increased learner satisfaction and more effective learning. Easier visualization was enabled through 3-D imagery, which helped to reduce cognitive load, while interactivity was enabled through engagement with the imagery or others in team training. Balian et al. [11] exemplify this, whereby half of the participants delivered 80% of CPR compressions correctly due to a realistic simulation that gave auditory (heartbeat metronome) and visual (blood flow to organs) learner feedback (see Fig 1).

We found that fully immersive AR or VR can trigger feelings of 'deep immersion' where learners perceive to be transported into the learning environment. This was found to improve learning and also skill performance. Additionally, when transferable skills or knowledge are taught using VR or AR, skills can be enhanced and practiced in a safe learning environment. This results in knowledge and skill transfer to clinical practice. For example, when using VR to reduce occupational needlestick injury in medical/nurse trainees, only 31-35% experienced an injury during the first two months of the clinical internship, compared to 80% of past trainees [12].

Lastly, for novice healthcare workers, VR and AR interventions enable repeated practice, which can lead to technical proficiency, skill acquisition and an improved performance. Repeated attempts

on XR simulators significantly improved skills. For example, the mean catheter insertion depth at first attempt on an AR haptic simulator was 5.30cm, which increased by 0.14cm on average at each repeated attempt ($p=0.013$) [13].

3.2 Barriers and Facilitators to Implementation

Costs (e.g., for purchasing equipment) and negative attitudes and experiences (i.e., cyber-sickness, fatigue and cognitive load during first-time use) were reported as barriers. Others related to developing the scenarios, logistics (e.g., storage and cleaning) and creating a curriculum. Facilitators were decreasing costs through commercialization, increasing the cost-effectiveness and access to training (e.g., for shift workers), leadership, collaboration and a cultural shift toward acceptance, as XR gains traction.

4 CONCLUSION

This is the first realist review to explore XR training in upskilling health workers. We recommend testing the mechanisms using hypothesis-driven approaches. Application of XR training within community and care settings warrants further investigation, given that this sample was not explored. Applications could also include simple behavioral skills such as medical waste disposal or hand hygiene for which XR exists [14]. Research should continue to explore implementation experiences. Aside from identifying areas for research, our theory can help to support the development and implementation of XR training for healthcare providers.

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